



# Standard Form

STANDARD FORM IS A CONVENIENT WAY OF WRITING VERY LARGE AND VERY SMALL NUMBERS.

## SEE ALSO

◀ 18–21 Multiplication

◀ 22–25 Division

◀ 36–39 Powers and roots

## Introducing standard form

Standard form makes very large or very small numbers easier to understand by showing them as a number multiplied by a power of 10. This is useful because the size of the power of 10 makes it possible to get an instant impression of how big the number really is.

this is the power of 10

$$4 \times 10^3$$

### ◀ Using standard form

This is how 4,000 is written as standard form—it shows that the decimal place for the number represented, 4,000, is 3 places to the right of 4.

## How to write a number in standard form

To write a number in standard form, work out how many places the decimal point must move to form a number between 1 and 10. If the number does not have a decimal point, add one after its final digit.

### ▷ Take a number

Standard form is usually used for very large or very small numbers.

### ▷ Add the decimal point

Identify the position of the decimal point if there is one. Add a decimal point at the end of the number, if it does not already have one.

### ▷ Move the decimal point

Move along the number and count how many places the decimal point must move to form a number between 1 and 10.

### ▷ Write as standard form

The number between 1 and 10 is multiplied by 10, and the small number, the “power” of 10, is found by counting how many places the decimal point moved to create the first number.

very large number

$$1,230,000$$


add decimal point

$$1,230,000.$$


6 5 4 3 2 1

$$1,230,000.$$

the decimal point moves 6 places to the left



the power is 6 because the decimal point moved six places; the power is positive because the decimal point moved to the left

$$1.23 \times 10^6$$

the first number must always be between 1 and 10

very small number

$$0.0006$$


decimal point is already here

$$0.0006$$


1 2 3 4

$$0.0006$$

the decimal point moves 4 places to the right



the power is negative because the decimal point moved to the right

$$6 \times 10^{-4}$$

the power is 4 because the decimal point moved four places